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OPTIMAL MONETARY POLICY IN A MODEL OF THE CREDIT CHANNEL*

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We study a simple extension of the basic new-Keynesian set-up in which we relax the assumption of frictionless financial markets: due to asymmetric information and default risk, bank lending rates incorporate a spread over the deposit rate. We demonstrate that financial frictions affect aggregate dynamics mainly through their impact on firms' financing costs, which increase in both the deposit rate and in the spread between lending and deposit rates. Welfare includes a concern for smoothing these two financial market variables. Our numerical simulations suggest that an aggressive easing of policy is optimal in response to adverse financial market shocks.

The financial crisis of 2007–9 has led economists to revisit the relevance of financial market conditions for real economic activity.

At the end of 2007, following the deterioration in the performance of non-prime mortgages in the US, the interest rate spread between short-term uncollateralised and collateralised interbank loans increased sharply. This increase was quickly passed on to lending rates to non-financial firms and households and was arguably responsible for the deep recessions of 2008–9. Concerns about the adverse macroeconomic repercussions of deteriorating financial market conditions led to sharp cuts in monetary policy interest rates around the globe.

Through which channels can shocks arising in financial markets be transmitted to the real economy? Should financial market conditions matter *per se* for monetary policy, or should they only be taken into account to the extent that they affect output and inflation? Can an increase in credit spreads generate a large enough economic reaction to justify interest rate cuts of the magnitude observed at the peak of the 2007–9 financial crisis?

We answer these questions using a model that recognises the existence of financial market imperfections in a tractable way. We extend the basic new-Keynesian (henceforth, NK) set-up to include two features. First, we assume that firms need to pay wages in advance of production, which generates a need for external finance. Second, we assume asymmetric information between banks and firms and adopt the costly state verification set-up of Townsend (1979). Under these assumptions, debt arises as the

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optimal financial contract between banks and firms, and firms must borrow at a premium over the risk-free rate.

We obtain three main sets of results, which allow us to provide answers to the three questions outlined above.

First, we show that the log-linear approximation of the equilibrium relationships in our model is similar in structure to the one arising in the NK set-up. As in the NK case, private sector decisions can be characterised by an intertemporal IS equation and a Phillips curve. These relationships, however, include additional terms to reflect the existence of informational asymmetries. The answer to our first question is that financial market conditions matter for macroeconomic outcomes because they affect firms' marginal costs. More specifically, higher credit spreads lead to an increase in lending rates and thus in the marginal cost of credit for firms. This increase in marginal costs is passed on to final prices and, as a result, it affects output. The log-linearised equilibrium equations also show that technology and financial shocks generate inefficient fluctuations in the economy. This is striking in the case of technology shocks, because they produce fully efficient economic fluctuations in the model with frictionless financial markets. The difference is due to the fact that in the presence of credit frictions, technology shocks lead to variations in firms' exposure to external finance and leverage, generating inefficient volatility in credit spreads and bankruptcy rates.

Our second set of results concerns optimal policy. Using an analytic, second-order approximation of the welfare function, we demonstrate that welfare is directly affected not just by the volatility of inflation and the output gap, as in the benchmark case with frictionless financial markets, but also by the volatility of the nominal interest rate and of credit spreads. The answer to our second question is that, in general, financial market conditions matter *per se* for monetary policy. Nevertheless, the weight of financial market conditions on welfare appears to be small from a quantitative perspective. In our model, taking financial conditions into account solely to the extent that they affect output and inflation would produce similar optimal responses to non-financial shocks.

Our final set of results is related to the optimal response to financial shocks under commitment. In reaction to a negative shock to firms' net worth, which increases the credit spread and the overall cost of financing, a sharp cut in policy interest rates is warranted. The policy easing dampens the inefficient increase in markups which firms implement to cope with the higher cost of credit. As a result, the fall in real wages is cushioned and the drop in households' consumption is smoothed. The answer to our third question is therefore that an adverse financial market shock does warrant a decisive easing of monetary policy.

The benefits of an optimal policy response are clear when these outcomes are compared with the case in which policy follows a simple Taylor rule. Since the higher credit spread has a cost-push effect on inflation, a Taylor rule would prescribe an interest rate tightening in response to the financial shock. The adverse implications of the shock on markups, real wages and consumption would be exacerbated.

The optimal reaction to technology shocks in our model is not dramatically different from the case with frictionless financial markets and from the prescriptions of a simple policy rule of the Taylor type. More specifically, near complete inflation stabilisation remains optimal.

Our article is not the first attempt to analyse monetary policy in models with credit frictions. Ravenna and Walsh (2006) characterises optimal monetary policy when firms need to borrow in advance to finance production. However, there is no default risk in that model and the cost of financing for firms is the risk-free rate. We show that our model nests that of Ravenna and Walsh (2006) in the special case in which the costs of asymmetric information disappear. Faia and Monacelli (2007) compare the welfare losses of various optimised simple interest rate rules in models with a structure similar to ours, but it does not characterise fully optimal (Ramsey) monetary policy. Similarly, Christiano *et al.* (2008) argues that the monetary policy reaction to a stock market boom/bust cycle would be superior, in terms of welfare, if liquidity developments were taken into account in a simple rule.

Our article is closest to Cúrdia and Woodford (2010), which also characterises optimal monetary policy in a simple framework where financial market imperfections matter. However, in our article, credit frictions arise because of heterogeneity and asymmetric information in firms' productivity, while in Cúrdia and Woodford (2010) they arise because of heterogeneity in households' spending opportunities. A second difference is that we characterise an optimal financial contract explicitly, while Cúrdia and Woodford (2010) rely on a reduced-form intermediation technology.

Demirel (2009) also studies optimal monetary policy in a model with microfounded financial frictions similar to ours but does not consider financial shocks. His results can therefore not be used to provide an answer to the questions we ask in this article.

The article proceeds as follows. In Section 1, we describe the environment and present the reduced-form system of linearised equilibrium conditions. This enables us to highlight the effect of financial market frictions on inflation and output dynamics and to compare our set-up to the basic NK model. Section 2 presents new empirical evidence on the Phillips curve and shows that the joint hypothesis that the cost channel and the credit channel are relevant for firms' price setting cannot be rejected. We outline the quantitative properties of our model in Section 3, under the assumption that monetary policy follows a simple rule. We present impulse responses to technology and monetary policy shocks and compare them to those arising under the model of the cost channel and the NK model. In Section 4, we derive a quadratic approximation of the social welfare. In Section 5, we present two extensions of the model, which we use later on to check the robustness of our results to key assumptions. In Section 6, we characterise numerically the features of optimal monetary policy under commitment. Section 7 concludes.[1]

1. The Environment

The economy is inhabited by a representative infinitely lived household and by a continuum of risk-neutral entrepreneurs. Households own firms producing differentiated goods in the retail sector, while entrepreneurs own firms producing a homogeneous good in the wholesale sector.

[1] In the online Appendix, we provide details of the financial contract, the price setting, the log-linearised system of equilibrium conditions in reduced form, the system in terms of gaps from the efficient equilibrium, the model with zero monitoring costs, the second-order welfare approximation and the two extensions of the model used in our robustness analysis.

Financial market imperfections, in the form of asymmetric information and costly state verification, affect the activity of wholesale firms. These firms produce according to a technology that is linear in labour and subject to idiosyncratic productivity shocks. Entrepreneurs need to raise external finance to pay workers in advance of production but, due to the idiosyncratic shock, they face default risk on their debt. Lending occurs through perfectly competitive financial intermediaries, which are able to ensure a safe return to households by providing funds to the continuum of firms. Firms and banks stipulate debt contracts, which are the optimal contractual arrangements between lenders and borrowers in this costly state verification environment.

The timing of events is as follows. At the beginning of the period, after the occurrence of aggregate shocks, the financial market opens. Households decide how to allocate nominal wealth among existing assets, namely money, a portfolio of nominal state-contingent bonds and deposits. Deposits are collected by a bank and used to finance firms' production. In the second part of the period, the goods market opens. Wholesale firms produce homogenous goods and sell them to the retail sector. If revenues are sufficient, they repay the debt and devote remaining profits to finance entrepreneurial consumption. Otherwise, they default and their production is seized by banks. Firms in the retail sector use the homogeneous good as input in the production of differentiated goods. Because of this product differentiation, retail firms have market power and become price makers. However, they are not free to change their price at will, because prices are subject to Calvo contracts. Retail goods are then purchased by households and wholesale entrepreneurs for own consumption.

1.1. *Households*

At the beginning of period t, the financial market opens. First, interest on financial assets acquired at time $t-1$ is paid. The households, holding an amount W_t of nominal wealth, allocate it among nominal assets, namely money M_t, a portfolio of state-contingent bonds Z_{t+1}, each paying a unit of currency in a particular state in period $t+1$, and one-period deposits denominated in units of currency D_t paying back $R_t^d D_t$ at the end of the period.

In the second part of the period, the goods market opens. Households' money balances are increased by the nominal amount of their revenues and decreased by the value of their expenses. Taxes are also paid or transfers received. The amount of nominal balances brought into period $t+1$ is equal to $M_t + P_t w_t h_t + V_t - P_t c_t - T_t$, where h_t is hours worked, w_t is the real wage, V_t is nominal profits transferred from retail producers to households and T_t is lump-sum nominal taxes collected by the government. c_t denotes a constant elasticity of substitution (CES) aggregator of a continuum $j \in (0,1)$ of differentiated consumption goods produced by retail firms, $c_t = [\int_0^1 c_t(j)^{\frac{\varepsilon-1}{\varepsilon}} \mathrm{d}j]^{\frac{\varepsilon}{\varepsilon-1}}$, with $\varepsilon > 1$. $P_t(j)$ denotes the price of good j, and $P_t = [\int_0^1 P_t(j)^{1-\varepsilon} \mathrm{d}j]^{\frac{1}{1-\varepsilon}}$ is the price of the CES aggregator. Nominal wealth at the beginning of period $t+1$ is given by

$$W_{t+1} = Z_{t+1} + R_t^d D_t + R_t^m (M_t + P_t w_t h_t + V_t - P_t c_t - T_t), \tag{1}$$

where R_t^m denotes the interest paid on money holdings.

The household's problem is to maximise preferences, defined as

$$\mathrm{E}_o\left\{\sum_{0}^{\infty}\beta^t[u(c_t)+\kappa(m_t)-v(h_t)]\right\},$$

where $u_c > 0$, $u_{cc} < 0$, $\kappa_m \geq 0, \kappa_{mm} < 0$ and $v_h > 0$, $v_{hh} > 0$ and $m_t \equiv M_t/P_t$ denotes real balances, subject to the budget constraint

$$M_t + D_t + E_t(Q_{t,t+1}Z_{t+1}) \leq W_t. \tag{2}$$

1.2. *Wholesale Firms*

The wholesale sector consists of a continuum of competitive firms, indexed by i, owned by entrepreneurs. Each firm produces the amount $y_{i,t}$ of a homogeneous good, using the technology

$$y_{i,t} = A_t\omega_{i,t}l_{i,t}, \tag{3}$$

where $l_{i,t}$ is the firm-specific labour input. A_t is an aggregate, serially correlated productivity shock and $\omega_{i,t}$ is an idiosyncratic, iid productivity shock with distribution function Φ and density function ϕ.

The production function (3) reflects our choice to abstract from capital accumulation. This is in contrast with most of the literature that introduces credit frictions in macro-models, where entrepreneurs are assumed to decide in period t how to allocate their profits to consumption and investment expenditures (see Carlstrom and Fuerst, 1997; Bernanke *et al.*, 1999). The value of the stock of capital available to firms in period $t+1$ provides the firm with a certain net worth (internal funds) that can be used in that period production. In that environment, aggregate shocks affect the evolution of firms' net worth, thus creating endogenous persistence. In our model, we assume instead that each firm receives a stochastic endowment τ_t at the beginning of each period, which can be used as internal funds. As these funds are not sufficient to finance the firm's desired level of production, firms need to raise external finance. As a result, financial frictions have important effects also in our economy. For example, a spread arises endogenously between the loan rate and the risk-free rate, to reflect the existence of default risk. At the same time, our simpler set-up enables us to provide an analytical characterisation of economic dynamics and of optimal policy in the presence of credit constraints and information asymmetry.

1.2.1. *Labour demand*

As in Christiano and Eichenbaum (1992) and Ravenna and Walsh (2006), we assume that firms need to raise external finance to pay for production inputs. Before observing the idiosyncratic productivity shock and after observing the aggregate shocks, they raise credit in the amount $P_t(x_{i,t} - \tau_t)$, for total funds at hand $P_tx_{i,t}$,[2] where

[2] We limit the support of the aggregate productivity shock, A_t, to ensure that there is always a need for external finance. In the absence of this assumption, for sufficiently large negative shocks, $w_tl_{i,t}$ might be smaller than τ_t, in which case firms could pay the wage bill using only their nominal internal funds.

$$x_{i,t} \geq w_t l_{i,t}. \tag{4}$$

Entrepreneurs sell output only to retailers. Let $\bar{P}_t$ be the price of the wholesale homogeneous good, and $\bar{P}_t/P_t = \chi_t^{-1}$ the relative price of wholesale goods to the aggregate price of retail goods. Each firm i maximises expected profits, subject to the financing constraint (4). Optimality requires that (4) holds with equality and

$$q_{i,t} = q_t = \frac{A_t}{w_t \chi_t}, \tag{5}$$

$$\mathcal{E}(y_{i,t}) = \chi_t q_t x_{i,t}, \tag{6}$$

where $q_{i,t} - 1$ is the Lagrange multiplier on the financing constraint and the expectation $\mathcal{E}[\cdot]$ is taken with respect to the idiosyncratic shock unknown at the time of labour hiring decision.

Equation (6) states that as the production function is constant return to scale, wholesale firms must sell at a markup $\chi_t q_t$ over firms' production costs. This allows them to cover for monitoring costs and the monopolistic distortion in the retail sector.

1.2.2. *The financial contract*

Loans are stipulated in units of currency after all aggregate shocks have occurred and repaid at the end of the same period. Lending occurs through the financial intermediary, which collects deposits from households and uses them to finance loans to firms.

Firms face an idiosyncratic productivity shock, whose realisation is observed at no costs only by the entrepreneur. The financial intermediary can monitor its realisation but only at a cost, which is a fraction μ of the value of the funds, $P_t x_{i,t}$. If the realisation of the idiosyncratic shock is sufficiently low, the value of the firm's production is not sufficient to repay the loan and the firm defaults, in which case its production is completely seized by the intermediary.

Define $f(\bar{\omega}) \equiv \int_{\bar{\omega}}^{\infty} \omega\Phi(\mathrm{d}\omega) - \bar{\omega}[1 - \Phi(\bar{\omega})]$ and $g(\bar{\omega}) \equiv \int_0^{\bar{\omega}} \omega\Phi(\mathrm{d}\omega) - \mu\Phi(\bar{\omega}) + \bar{\omega}[1 - \Phi(\bar{\omega})]$ as the expected shares of output accruing, respectively, to an entrepreneur and to a lender, after stipulating a contract that sets the fixed repayment at $\bar{P}_t \chi_t q_t \bar{\omega}_{it} x_{i,t}$ units of money. $\bar{\omega}_t$ is a threshold for the distribution of the idiosyncratic productivity shock below which firms default. At the individual firm level, total output is split between the entrepreneur, the lender and monitoring costs so that $f(\bar{\omega}_t) + g(\bar{\omega}_t) = 1 - \mu\Phi(\bar{\omega}_t)$.

The informational structure corresponds to a costly state verification problem. The solution is a standard debt contract (see Gale and Hellwig, 1985) which is derived in Section A of the online Appendix. The terms of the contract are identical for all firms. The optimality conditions can be written as

$$q_t = \frac{R_t}{1 - \mu\Phi(\bar{\omega}_t) + \dfrac{\mu f(\bar{\omega}_t)\phi(\bar{\omega}_t)}{f_{\bar{\omega}}(\bar{\omega}_t)}}, \tag{7}$$

$$x_t = \frac{R_t \tau_t}{R_t - q_t g(\bar{\omega}_t)}. \tag{8}$$

Compared with the standard assumption of real debt contracts employed by Carlstrom and Fuerst (1997, 1998) and Bernanke *et al.* (1999), our assumption of nominal contracts has two consequences.

The first is that monetary policy has real effects in our model – beyond those caused by the assumption of Calvo prices. These effects arise mainly through the impact of the nominal interest rate on the financial markup q_t. An increase in the nominal interest rate increases the opportunity cost of lending funds for the financial intermediary and is therefore passed on to loan rates. Thus, the real effects in our model are similar to those present in a cost-channel model. Loan rates, however, increase more than one-to-one with respect to the risk-free rate. The increase in the latter variable makes it more difficult for firms to pay back their debt, and default probabilities must increase. As a result, credit spreads must also rise in equilibrium, increasing the average financial distortion and the markup q_t.

The second effect of the assumption of nominal contracts is that the fraction of the loan lost in monitoring activities is also in terms of currency, not in terms of physical goods – as is typically the case when contracts are in real terms. Intermediate firms sell their entire output to the retail sector at the end of the period and use the monetary proceedings from the sale to pay bank loans. To the extent that banks choose to monitor individual firms' productivity levels, some of the money will not be available to pay households' deposits. Thus, financial frictions do not generate a loss of resources in our economy but introduce an additional cost to be taken into account by banks when agreeing on an appropriate interest rate on loans. An important implication of this assumption is that fluctuations in bankruptcy rates will only have an impact on welfare indirectly, to the extent that they have undesirable implications on the markup q_t or in the amount of loans. With real contracts, on the contrary, monitoring costs amount to a destruction of goods which would otherwise have been available for consumption: fluctuations in bankruptcy rates therefore have a direct welfare cost.

The gross interest rate on loans can be backed out from the debt repayment, which requires $\bar{P}_t\bar{\omega}_t\chi_t q_t x_t = R_t^l P_t(x_t - \tau_t)$. This expression can be used to write the spread between the loan rate and the risk-free rate, $\Delta_t \equiv R_t^l/R_t^d$, as

$$\Delta_t = \frac{\bar{\omega}_t}{g(\bar{\omega}_t)}. \tag{9}$$

1.2.3. *Entrepreneurs*

Each entrepreneur i has linear preferences over consumption and is infinitely lived. He or she discounts the future at the same rate as the households, β, and consumes a CES basket of differentiated goods, $e_{i,t} = [\int_0^1 e_{i,t}(j)^{\frac{\varepsilon-1}{\varepsilon}}\mathrm{d}j]^{\frac{\varepsilon}{\varepsilon-1}}$, where $e_{i,t}(j)$ is firm i's consumption of good j. At the end of each period, each entrepreneur sells his or her output to the retail sector and, if feasible, repays the debt. Remaining profits are entirely allocated to final consumption goods.

Aggregating across firms, we obtain $e_t = f(\bar{\omega}_t)q_t x_t$, where $e_t = \int_0^1 e_{i,t}\mathrm{d}i$ is the aggregate entrepreneurial consumption of the final consumption good. Using (7)–(8), we can rewrite aggregate entrepreneurial consumption as

$$e_t = \tau_t R_t \left[1 + \frac{\mu\phi(\bar{\omega}_t)}{f_{\bar{\omega}}(\bar{\omega}_t)}\right]^{-1}. \tag{10}$$

Equation (10) shows that entrepreneurial consumption depends only on the nominal interest rate, on the bankruptcy threshold $\bar{\omega}_t$, and on the exogenous shock τ_t. As mentioned above, an increase in the nominal interest rate increases the markup q_t, which reflects into higher firms' profits and consumption. Entrepreneurial consumption increases also when firms' internal funds τ_t rise. This reduces leverage and the spread, increasing firms' profits. Changes in the threshold $\bar{\omega}_t$ act instead by modifying the output share $f(\bar{\omega}_t)$ (together with $g(\bar{\omega}_t)$). As $f(\bar{\omega}_t)$ and entrepreneurs' profits are decreasing in the threshold, an increase in bankruptcy rates tends to depress entrepreneurial consumption.

1.3. *Retail Firms*

As in Bernanke *et al.* (1999), a continuum of monopolistically competitive retailers buy wholesale output from entrepreneurs in a competitive market and then differentiate it at no cost. Profits are distributed to the households, who own firms in the retail sector.

Let $Y_t(j)$ be the quantity of output sold by retailer j. This quantity can be used for households' consumption, $c_t(j)$, and for entrepreneurs' consumption, $e_t(j)$. The final good Y_t is the same CES composite of individual retail goods as c_t.

We assume that each retailer can change its price with probability $1 - \theta$, following Calvo (1983). Price setting is thus standard and is characterised in online Appendix B. We define the relative price of differentiated good j as $p_t(j) \equiv P_t(j)/P_t$ and the relative price dispersion term as $s_t \equiv \int_0^1 (p_t(j))^{-\varepsilon} \mathrm{d}j$.

1.4. *Monetary Policy*

Monetary policy will be characterised either as an optimal Ramsey plan or as a simple Taylor-type rule. In addition, the central bank needs to specify a rule for either R_t^m or M_t^s. It is convenient to express this rule in terms of R_t^m, or alternatively in terms of $\Delta_{m,t} \equiv (R_t - R_t^m)/R_t$.

In order to facilitate the comparison of our model with the NK set-up, we assume that $\Delta_{m,t} = \Delta_m$, for all i. Under this policy, money demand becomes recursive and can be neglected for the solution of the system. Moreover, the households' optimality conditions are identical to those obtained in the benchmark NK model.[3]

[3] Money has nonetheless an important role in our model, as it makes credit possible. External finance arises because workers will not supply labour on credit but demand advance payment. Wage payments are made in the financial market, which opens at the beginning of the period and are then used to finance expenditures in the goods market at the end of the period. Other nominal assets, deposits and bonds have a one-period maturity and therefore cannot be used for this purpose. Under our timing, as in Lucas and Stockey (1987), money would affect the equilibrium. To facilitate the comparison with the standard new-Keynesian model, we neutralise this effect by assuming that money is remunerated at a rate that is proportional to the risk-free rate.

1.5. *The Linearised Equilibrium Conditions*

In order to characterise the optimal response of monetary policy, we first log-linearise the non-linear system of equilibrium conditions around a zero-inflation steady state (characterised in online Appendix C). We then express these conditions in deviations from the efficient equilibrium, that is, an equilibrium where all (nominal and real) financial frictions are removed, and prices are fully flexible. This latter is characterised in online Appendix D.

We assume a functional form $U(c_t; h_t) - v(h_t) = c_t^{1-\sigma^{-1}}/(1-\sigma^{-1}) - \psi h_t^{1+\varphi}/(1+\varphi)$. Define $\pi_{t+1} \equiv \log(P_{t+1}/P_t)$, $\hat{p}_t(j) = \log p_t(j)$, $a_t = \log A_t$ and $\hat{\tau}_t = \log \tau_t$. We find it useful to define the output gap, $\tilde{Y}_t$, as actual output in deviation from efficient output, where each variable is linearised around its steady state (i.e. Y for actual output, Y^e for efficient output). Under this definition, there is a non-zero steady-state output gap $\log Y - \log Y^e$ when $\tilde{Y}_t = 0$. The system can be written as

$$\delta_1 \hat{\Delta}_t = \left(1 + \varphi + \sigma^{-1}\frac{Y}{c}\right)\tilde{Y}_t - \sigma^{-1}\frac{e}{c}\hat{R}_t + \xi_{1,t}, \tag{11}$$

$$\begin{aligned}\tilde{Y}_t = {} & \mathrm{E}_t\tilde{Y}_{t+1} - \sigma\left(\frac{1+\sigma^{-1}\frac{e}{c}}{1-\varphi\frac{e}{c}}\right)\left(\hat{R}_t - \mathrm{E}_t\pi_{t+1}\right) \\ & - \left(\frac{\alpha_1 - \alpha_2\frac{e}{c}}{1-\varphi\frac{e}{c}}\right)\left(\hat{\Delta}_t - \mathrm{E}_t\hat{\Delta}_{t+1}\right) + \frac{\frac{e}{c}}{1-\varphi\frac{e}{c}}\left(\hat{R}_t - \mathrm{E}_t\hat{R}_{t+1}\right) + \xi_{2,t},\end{aligned} \tag{12}$$

$$\pi_t = \bar{\kappa}[(\sigma^{-1}\alpha_1 + \alpha_2)\hat{\Delta}_t + (\sigma^{-1} + \varphi)\tilde{Y}_t + \hat{R}_t + \xi_{3,t}] + \beta\mathrm{E}_t\pi_{t+1}, \tag{13}$$

for coefficients α_1, α_2 and δ_1 defined in online Appendix D. Note that $\bar{\kappa} \equiv (1+\sigma^{-1}e/c)^{-1}\ (1-\theta)(1-\beta\theta)/\theta$, $\alpha_1 > 0$ and $\alpha_2 > 0$. Under our parameterisation (described below) also $\delta_1 > 0$. The term in brackets multiplying the change in the spread in the IS curve is also positive, $(\alpha_1 - \alpha_2 e/c)/(1 - \varphi e/c) > 0$.

Equation (11) shows that the spread between the loan rate and the policy rate increases with excess aggregate demand, $\tilde{Y}_t$. An increase in the demand for retail (and thus also for wholesale) goods implies an implicit tightening of the credit constraint, as the given amount of internal funds must be used to finance a higher level of debt. The increased default risk generates a larger spread. For the same reasons, the spread decreases with the amount of internal funds, $\hat{\tau}_t$. An increase in the nominal interest rate, $\hat{R}_t$, generates a reduction in the demand for final goods and thus in the demand for input in their production (wholesale goods). For a given amount of internal funds, leverage and the risk of default fall, reducing the spread.

Equation (12) is a forward-looking IS-curve describing the determinants of the output gap. As in the standard NK model, the gap is affected positively by its expected future value and negatively by the real interest rate. In our model, however, the output gap also depends on the expected change in the nominal interest rate and in the credit spread. Note that this dependence is not present in a cost-channel model: it would disappear in the absence of monitoring costs. A higher credit spread is contractionary in our model, because it induces an increase in bankruptcy rates and a fall in entrepreneurial consumption. In our calibration, an expected increase in the spread between periods t and $t+1$ tends instead to be expansionary, in spite of the fact that entrepreneurs are myopic in their consumption patterns. Through the aggregate resource constraint, the reduction in $t+1$ entrepreneurial consumption, which is due to the higher expected spread, also tends to imply an increase in future households' consumption. As households are forward looking, this effect will feed through to current households' consumption, thereby leading to an expansionary effect on output. On top of the standard real interest rate effect, changes in the nominal interest rate have an additional impact on output which operates through a similar channel. *Ceteris paribus*, a higher nominal interest rate will have a small expansionary effect, as it will increase the financial markup and entrepreneurial consumption. However, an expected increase in the nominal interest rate will be contractionary, as it will lead to an expected fall in households' future consumption.

Equation (13) represents an extended Phillips curve. The first determinant of inflation is the output gap. This term is standard, even if it enters with a different coefficient reflecting the presence of entrepreneurs in the economy. As in the cost-channel model, (13) also includes a nominal interest rate term, whose increase pushes up marginal costs. Finally, the novel feature of our model is the presence of a credit spread in the equation. A higher credit spread implies a higher cost of external finance for wholesale firms and therefore exerts independent pressure on inflation.

The credit spread and the nominal interest rate act as endogenous 'cost-push' terms in the economy. While pushing up marginal costs and inflation, an increase in either term also exerts downward pressure on economic activity. For the nominal interest rate, the main channel of transmission to aggregate demand works through the effect of the increase in the real interest rate, which induces households to postpone their consumption to the future. For the credit spread, transmission mainly occurs through an increase in the financial markup and a fall in the real wage, which enable firms to offset the increase in financing costs.

All exogenous disturbances lead to inefficient fluctuations. For instance, the output expansion, which will typically follow a positive technology shock, generates the need for an increase in external finance and in leverage, hence leading to an increase in the credit spread, marginal costs and inflation. In turn, the higher credit spread will have a contractionary effect on output through the channel described above. Similarly, a reduction in firms' internal funds leads to an increase in the credit spread and inflation, while at the same time exerting a contractionary effect on the output gap through the IS curve.

Our model nests both the cost-channel model of Ravenna and Walsh (2006) and the standard NK model. Consider first the special case when monitoring costs are zero, that is, $\mu = 0$, and entrepreneurs' net worth is zero, that is, $\tau_t \approx 0$, for all t. In this case, firms still need to borrow in advance of production. However, the information asymmetry concerning wholesale firms' productivity disappears because banks can monitor at no cost. Economic dynamics is characterised (see online Appendix E) by the system of equilibrium conditions obtained by Ravenna and Walsh (2006) in their model of the cost channel, where firms borrow in advance of production but, as there is no asymmetric information nor default risk, they simply pay the risk-free rate on these funds. Finally, the system would boil down to the NK model in the absence of nominal debt contracts, in which case the nominal interest rate would not affect marginal costs.

2. Empirical Evidence

In line with existing literature, we calibrate the structural parameters to match unconditional data moments. We set long-run monitoring costs at 12% of firms' output, that is, $\mu = 0.12$, a value consistent with the empirical estimates in Levin *et al.* (2004). We then assume that idiosyncratic shocks are lognormal and calibrate their standard deviation (σ_ω) and the subsidy τ so that that the annualised steady-state spread Δ is approximately equal to 2% and roughly 1% of firms go bankrupt each quarter. As to monopolistic competition and retail pricing, we assume $\varepsilon = 11$, leading to a steady-state markup of 10%, and a probability of not being able to re-optimise prices $\theta = 0.66$, implying that prices are changed on average every three quarters. For preferences, we adopt the calibration in Woodford (2003), which implies $\varphi = 0.11$ and $\sigma = 0.16$.[4] Finally, we set the persistence of technology shocks to 0.9. For firms' net worth shocks, we set the autocorrelation coefficient to 0.7317, so as to obtain the same persistence of net worth as in the case where it evolves endogenously (see the model extensions presented in Section 5).

These parameter values will not only have implications for the steady state but also on the coefficients of the linearised equations. One of the advantages of our simple model is to make these implications very transparent, due to the analytical form of the linearised model.

For example, our calibration is consistent with a very small elasticity of the credit spread to the output gap $\left((1 + \varphi + \sigma^{-1} Y/c)/\delta_1 = 0.015\right)$ and especially to the deposit rate $\left(\sigma^{-1}(e/c)/\delta_1 = 0.0003\right)$ – see (11). This suggests that the credit spread should be relatively insensitive to mild economic fluctuations.

Concerning the IS curve (12), the quantitative relevance of the terms which differentiate it from the standard New-Keynesian version is very sensitive to the steady-state levels of entrepreneurs' and households' consumptions (the e/c term). These variables are hard to calibrate directly, given the difficulty of finding empirical counterparts to the model's notions of households and entrepreneurs.

A much more robust, testable implication of our model concerns the presence of credit spreads in the Phillips curve. From (14), it is immediately clear that their presence is robust to the $e \to 0$ case. Our benchmark calibration has the following implication for the coefficient of the spread in the Phillips curve: $\sigma^{-1}\alpha_1 + \alpha_2 = 6.0$.

This coefficient is strikingly large and has important implications. It suggests that a 1% point increase in the loan rate $\hat{R}^l_t = \hat{R}_t + \hat{\Delta}_t$ has different effects depending on its sources. *Ceteris paribus*, an increase in the credit spread component produces six times larger impact on marginal costs than an increase in the deposit rate $\hat{R}_t$ of the same size. In the remainder of this Section, we examine the empirical plausibility of this result and provide an economic intuition.

[4] De Fiore and Tristani (2009) present the result of an alternative calibration where $\varphi = 0$ and $\sigma = 1$. The results are qualitatively similar.

We estimate the Phillips curve using the generalised method of moments (GMM) methodology applied by Galí and Gertler (1999) and Ravenna and Walsh (2006).[5] GMM has the advantage of allowing us to focus solely on the equation where financial frictions have the most significant impact, the Phillips curve, without the need to take a stand on the entire structure of the economy or on the distribution of the shocks – see also the discussion in Galí *et al.* (2005). Ravenna and Walsh (2006) test the cost-channel hypothesis using data over the period 1960:Q1–2001:Q1 and conclude that the hypothesis cannot be rejected. We test jointly for the presence of the cost channel and the credit channel in the new-Phillips curve. As a first step, we rewrite the Phillips curve in terms of real marginal costs averaged across all firms. We then show that this formulation is robust to a more realistic version of our model – a version including capital – which is amenable to empirical analysis.

Denote average real marginal costs of wholesale goods as u_t. They are given by the ratio of average unit labour costs to the average marginal product of labour. In our model, $u_t = w_t h_t / y_t$. We can use firms' labour demand (5) to express the average real marginal costs of retail goods as $\chi_t^{-1} = u_t q_t$, and to rewrite the new-Phillips curve as

$$\hat{\pi}_t = \lambda(\hat{u}_t + \hat{R}_t + \alpha_2 \hat{\Lambda}_t) + \beta \mathrm{E}_t \hat{\pi}_{t+1}, \tag{14}$$

where we used $\hat{q}_t = \hat{R}_t + \alpha_2 \hat{\Lambda}_t$. Note that $\alpha_2 = 4.1$ in our calibration of the model. Also, in the efficient equilibrium, $u_t^e = w_t^e / A_t = 1$, implying that $\tilde{u}_t = \hat{u}_t$.

Consider now an extension of our model where the production function (3) is replaced by $y_{i,t} = A_t \omega_{i,t} l_{i,t}^a k_{i,t}^{1-a}$ and where firms need to finance the entire input bill – wage costs and the rental of capital – in advance of production. The financing constraint (4) becomes $x_{i,t} \leq w_t l_{i,t} + \rho_t k_{i,t}$. Average real marginal costs for retail firms are still given by χ_t^{-1} but firms' labour demand now implies that $\chi_t^{-1} = q_t u_t / a$. This latter condition differs from the one obtained in the absence of capital up to the labour share in output, a, which is a constant.

Finally, we use the correction derived by Sbordone (2002) to allow for firm-specific capital. In this case, inflation can be related to average real marginal cost through an equation similar to (14), with λ replaced by $\tilde{\lambda} \equiv \lambda/\zeta$, where ζ is a coefficient given by $\zeta = [1 - a(1 - \varepsilon)]/(1 - a)$.

On the basis of this formulation, we use data on unit labour costs to measure average real marginal costs $\hat{u}_t$. If we define as $\mathbf{z}_t$ a vector of variables within firms' information set that are orthogonal to their error in forecasting future inflation, the Phillips curve implies the orthogonality conditions

$$\mathrm{E}_t \left\{ \left[\theta \hat{\pi}_t - (1-\theta)(1-\theta\beta) \left(\frac{\hat{u}_t}{\zeta} + \eta \frac{\hat{R}_t}{\zeta} + \alpha_2 \frac{\hat{\Delta}_t}{\zeta} \right) - \theta\beta \hat{\pi}_{t+1} \right] \mathbf{z}_t \right\} = 0. \tag{15}$$

We calibrate ζ using a labour share $a = 2/3$ and, as already mentioned above, a value of 11 for ε – these are the same parameter values used in Galí and Gertler (1999), Galí *et al.* (2001) and Ravenna and Walsh (2006). We then estimate the remaining

[5] We wish to thank Federico Ravenna for providing us with data and code used in the Ravenna and Walsh (2006) analysis.

Table 1
Parameter Estimates

	θ	β	η	α_2	H_0: $\alpha_2 = 4.1$	Hansen test
Ravenna–Walsh instruments						
Sample: 1960:1–2007:2	0.586	0.926	1.005	5.959	0.545	12.81
	(0.03)	(0.03)	(0.52)	(2.52)	(0.46)	(0.99)
Restricted	0.585	0.926	1.0	5.870		12.83
	(0.02)	(0.03)	(-)	(2.14)		(0.99)
Extended sample: 1960:1–2010:1	0.648	0.949	1.469	12.822	5.819	13.255
	(0.03)	(0.03)	(0.72)	(3.62)	(0.02)	(0.99)
Ravenna–Walsh sample: 1960:1–2001:1	0.504	0.847	0.980	3.070	0.265	11.63
	(0.02)	(0.03)	(0.32)	(2.00)	(0.61)	(0.99)
Galí–Gertler instruments						
Sample: 1960:1–2007:2	0.657	0.965	1.549	8.122	0.994	10.797
	(0.06)	(0.03)	(1.03)	(4.03)	(0.32)	(0.63)

Notes. Two-stage GMM estimates of the structural parameters of (15). Newey–West (12 lags) HAC standard errors in brackets, p-values in square brackets. 'Ravenna–Walsh instruments' denotes the instruments used by Ravenna and Walsh (four lags of: non-farm business sector real unit labour cost, HP-filtered output gap, GDP deflator inflation, the Commodity Research Bureau (CRB) commodity price index inflation, 10-year Ű3-month US government bond spread, non-farm business sector hourly compensation inflation, three-month T-bill rate) plus four lags of the credit spread. 'Galí–Gertler instruments' denotes the instruments used by Galí and Gertler (two lags of non-farm business sector real unit labour cost, HP-filtered output gap, non-farm business sector hourly compensation inflation, and four lags of GDP deflator inflation and three-month T-bill rate) plus four lags of the credit spread. All data in common with Ravenna and Walsh are from the Bureau of Labor Statistics, the Bureau of Economic Analysis, the Federal Reserve System FRED database. The credit spread is the spread between Moody's Corporate AAA Yield and the 10-year Government bond yield.

parameters θ, β, η and α_2 on US data using a GMM estimator. Note that the cost-channel assumption implies $\eta = 1$. Our credit channel model additionally requires $\alpha_2 \simeq 4.1$. The standard new-Keynesian model is obtained when $\eta = \alpha_2 = 0$.

We use the same variables as in Ravenna and Walsh (2006), plus the spread between Moody's Corporate AAA Yield and the 10-year Government bond yield. Our estimates – with Newey–West corrected standard errors in parenthesis – are presented in Table 1.

The first row displays our benchmark results, which are based on the 1960:Q1–2007:Q2 sample and on the instruments used by Ravenna and Walsh (2006) extended to include four lags of the credit spread.[6] We end the sample in 2007:Q2 to abstract from potentially more non-linear dynamics induced by the financial crisis. Our estimates for the Calvo parameter θ and for the discount factor β are broadly in line with previous results in this literature. We find that Ravenna and Walsh (2006)'s results are robust to the longer sample period. The point estimate for the η coefficient is almost exactly 1, as assumed in the model, and borderline significant at the 5% level. The α_2 coefficient on the credit spread is around 6 and significantly different from zero. Hence, the large sensitivity of inflation to the credit spread predicted by our model appears to be borne by the data. The hypothesis that α_2 be exactly equal to 4.1, as

[6] The Ravenna and Walsh (2006) instruments include four lags of: unit labour costs, GDP deflator inflation, a commodity price index inflation, the term spread, the nominal interest rate, wage inflation and the output gap measured through the Hodrick–Prescott filter.

implied by our steady-state calibration, would not be rejected by a Wald test (see the column H_0: $\alpha_2 = 4.1$). If we impose this restriction, the estimates of the other parameters remain almost unchanged (see the second row in the Table). The over-identifying restrictions cannot be rejected by the Hansen test.

The next two rows in the Table test the robustness of our results to a longer sample including the financial crisis – 1960:Q1–2010:Q1 – and to a shorter sample corresponding to that used by Ravenna and Walsh (2006) – 1960:Q1–2001:Q1. The inclusion of the crisis years in the sample has little impact on the estimates of θ and β but tends to increase the relevance of financial frictions for inflation. Both the point estimates for η and α_2 increase, with the latter reaching the strikingly high and significant value of 12.8. Such a high value could arise in our model through a larger variance of idiosyncratic shocks σ_{ω_i}, which could be seen as a way of capturing the large increase in uncertainty that characterised the crisis period.

We also present our estimates over the Ravenna and Walsh (2006) sample to facilitate a comparison with their results. Compared with the second row in their Table 1, there are very small changes in the estimates of θ, β and η when we include credit spreads in the Phillips curve. Over this shorter sample period, however, the credit spread coefficient is smaller and less significant.

Finally, we check the robustness of our parameter estimates to a different choice of instruments. The last row in Table 1 presents estimates based on the smaller set of instruments used in Galí and Gertler (1999), again augmented to include four lags of the credit spread.[7] Once again, the estimates for θ and β are little affected. The point estimate for α_2, however, increases to 8.1 and remains significant. The point estimate for η also increases but becomes insignificant.

All in all, our GMM estimates suggest that a key theoretical implication of our model, namely the large sensitivity of inflation to credit spreads, is not rejected by the data. The point estimates for the α_2 coefficient vary somewhat across samples and changing the instruments set but they always remain high and notably much higher than the η coefficient. This suggests that an increase in loan rates due to a larger credit spread, for given risk-free interest rate, has different implications from an increase due to a higher risk-free rate, for given credit spread.

To understand this result, it is useful to notice that financial frictions affect marginal costs through their impact on the financial markup q_t – see (14) above. This markup, in turn, is given by (7), where we can replace $\bar{\omega}_t$ with Δ_t using equation (9). To a first-order approximation, (7) demonstrates that an increase in the safe interest rate $\hat{R}_t$ is associated with a one-to-one increase in $\hat{q}_t$. *Ceteris paribus*, that is, for given bankruptcy rates, a firm will simply pass on the increase in its own funding costs to final prices. In contrast, an increase in the credit spread is tantamount to an increase in $\hat{\bar{\omega}}_t$ and in bankruptcy rates. Each firm must now take into account that the higher $\hat{\Delta}_t$ is not only associated with an increase in its own funding costs but also with an aggregate increase in monitoring costs. The overall increase in marginal costs is therefore higher, and prices must increase more, than in the case of an increase in $\hat{R}_t$.

[7] These instruments include two lags of the unit labour costs, wage inflation and the output gap, and four lags of GDP deflator inflation and the nominal interest rate.

3. Impulse Responses

As a benchmark for comparison with the optimal policy case, we provide some evidence on the quantitative implications of the model through an impulse response analysis. For this purpose, we close the model with a simple monetary policy rule of the Taylor type:

$$R_t = -\ln\beta + 1.5\hat{\pi}_t + 0.5\tilde{Y}_t + u_t^p, \tag{16}$$

where u_t^p is a monetary policy shock. For illustrative purposes, we assume that policy shocks are serially correlated with an autoregressive coefficient of 0.5.

Figure 1 shows impulse responses to a negative 1% technology shock under the Taylor rule in our credit channel model (denoted with 'DT'), in the model of the cost channel ('CC') and in the standard NK model with frictionless financial markets ('NK').[8] The Figure shows that financial frictions, and the specific way those frictions are modelled, play an important role also in the transmission to the economy of a shock arising outside the financial sector.

Following the shock, in the NK model the real wage falls by less than 1% – the reduction that would occur in the efficient equilibrium. As a consequence, a small positive output gap ('*ygap*') arises, inducing a mild rise in marginal costs ('$-\chi$') and in

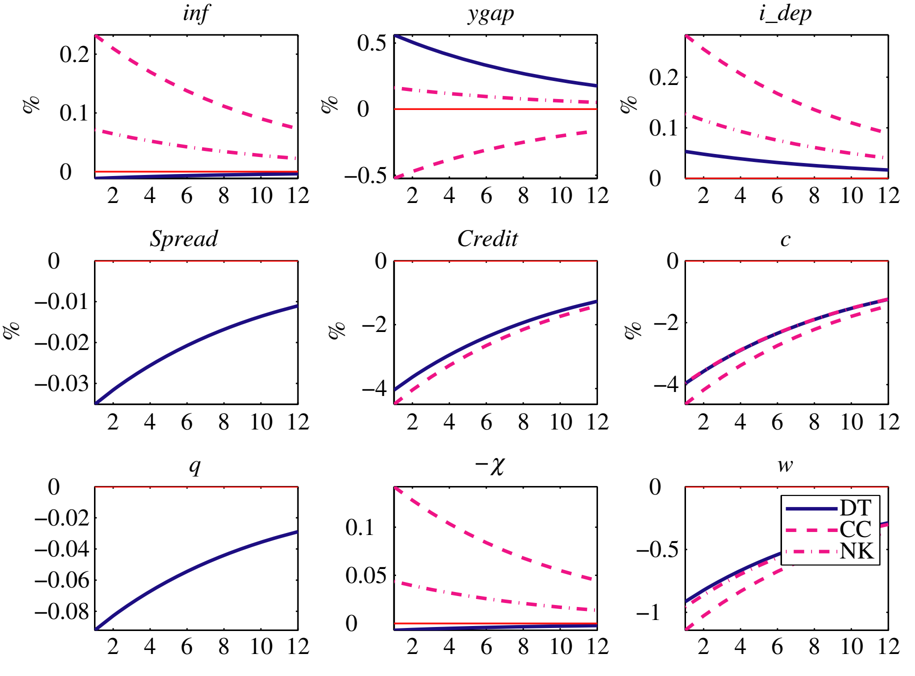

Fig. 1. *Impulse Responses to a Negative Technology Shock Under the Taylor Rule*
Notes: inf: Inflation Rate; *ygap*: Output Gap; *i dep*: Interest Rate on Deposits; *Spread*: Spread between the Loan Rate and the Deposit Rate; *Credit*: Credit Stock; *c*: Households' Consumption; *q*: Financial Markup; $-\chi$: Calvo Markup; *w*: Wage Rate.

[8] As the steady states of output y and of the efficient level of output y^e are different, the output gap term in the Taylor rule is written as $gap = \hat{y}_t - \hat{y}_t^e - y^*$.

inflation (‘*inf*’). The policy response is to increase the interest rate on deposits (‘*i_dep*’) before it slowly reverts to the steady state.

In the model of the cost channel, the behaviour of the output gap is qualitatively different. As the interest rate rises above the level observed in the NK model, marginal costs increase and the output gap turns negative throughout the entire adjustment period. The increase in marginal costs is also responsible for inflation remaining positive and above the path observed in the NK model.

Impulse responses in our model are in between the NK and cost-channel cases. The fall in output after the shock is accompanied by a reduction in the amount of credit extended to the economy. This implies a fall in leverage, as firms’ net worth is constant. As a result, the bankruptcy rate in the economy falls and so does the credit spread. The ensuing negative pressure on inflation dominates on the positive pressure arising from the higher interest rate and the positive output gap. Consequently, the response of inflation is slightly negative. The effect on the output gap is positive and close to the one observed in the NK model, although the mechanism is different. In our model, the fall in the spread and the increase in the interest rate generate an overall reduction in the financial distortion and in the financial markup, (‘*q*’) that firms need to charge. A lower markup limits the fall in real wages to below 1%, expanding household consumption. This effect explains about half of the positive response of the output gap. The fall in the spread also reduces the threshold for the idiosyncratic productivity shock below which firms default, raising entrepreneurs’ expected profits and aggregate entrepreneurial consumption. This latter effect accounts for the other half of the response of the output gap.

It should be emphasised that these specific results depend on the exact specification of the policy rule. With the Taylor rule estimated by Smets and Wouters (2007), which also includes an interest rate smoothing term, the responses of some variables – notably the output gap and inflation – to a technology shock would be closer across models. Nonetheless, Figure 1 highlights that the introduction of financial frictions can qualitatively alter the transmission of real shocks. The Figure also highlights the importance of agency costs. The output gap is positive in the NK model and negative in the model of the cost channel. In our model with both a cost channel and agency costs, under our parameterisation and policy rule, the output gap returns positive and above the one observed in the NK model. Agency costs – as reflected by the credit spread – more than offset the effect of the cost channel on marginal costs and the output gap.

In our model, following a technology shock, a positive relationship arises between the credit spread and output. A pro-cyclical response of the spread to this shock is standard in models adopting the Carlstrom and Fuerst (1997) set-up but the data show that spreads tend to increase during recessions – this is the case, for example, for the difference between lowest and highest rates on corporate bond yields in the US (see, e.g. Figure 1 in Levin *et al.*, 2004). Our model would indeed be capable of generating countercyclical credit spreads, if other shocks were allowed to drive business cycle fluctuations. For example, Figure 4 below shows that shocks to the value of firms’ net worth do give rise to a countercyclical response of the credit spread. A combination of technology shocks and shocks to firms’ net worth can generate a negative unconditional correlation between output and credit spreads.

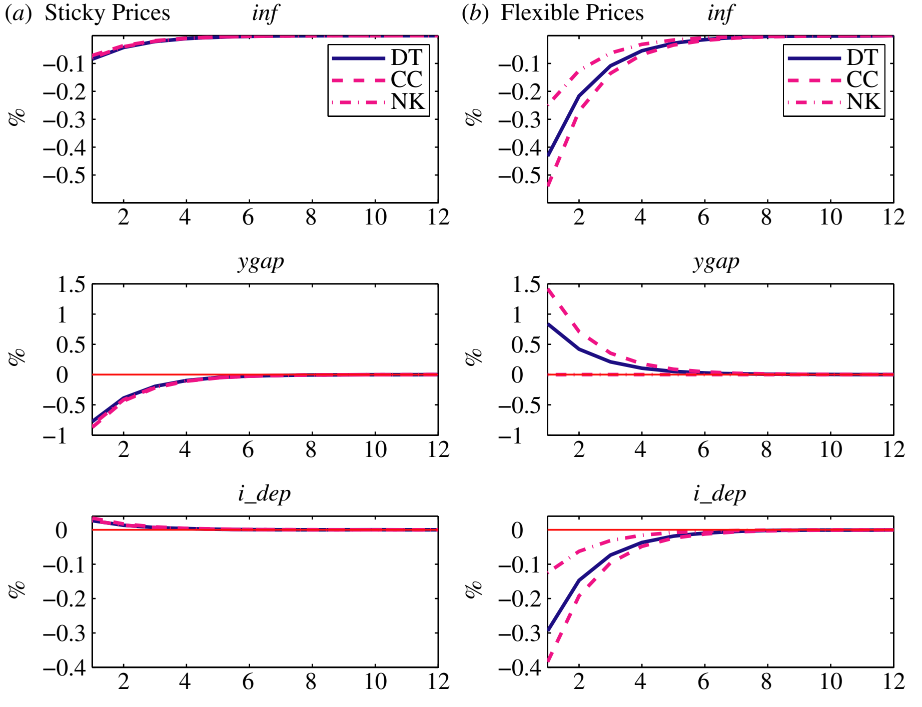

Fig. 2. *Impulse Responses to a Policy Shock* (see Notes to Figure 1)

Figure 2 shows impulse responses to a restrictive monetary policy shock when prices are either sticky or flexible.

Panel (*a*) plots the responses of inflation, output and the interest rate in the three different models considered above. Contrary to the case of a technology shock, the behaviour of inflation and output is very similar. In all models, the shock increases the nominal and real interest rate, depresses aggregate demand, lowering marginal costs and inflation. In the model of the credit channel, the ensuing reduction in credit and leverage lowers also the credit spread, leading overall to a fall in the financial markup.

Panel (*b*) shows the impulse responses of inflation, output and the interest rate in versions of the three models where price stickiness is removed. The plot shows that the presence of the cost channel and of agency costs creates real effects of monetary policy over and above those arising due to the presence of sticky prices. Because prices are flexible, the policy shock induces an increase of the price level on impact and a subsequent reduction of inflation. The policy rule implies an immediate reduction of the nominal interest rate. In the frictionless (NK) model, the real interest rate does not move, implying no effect of the monetary policy shock on the output gap. In the CC and DT models, the fall in the nominal interest rate has a direct effect on real marginal costs, generating a positive output gap.

Our analysis so far suggests that the introduction of credit frictions, and the way these frictions are modelled, has important implications for the transmission of non-financial shocks (e.g. technology shocks) to the real economy. It also suggests, however, that the existence of credit frictions is not a sufficient ingredient for financial variables to play a quantitatively important role in shaping the monetary policy transmission mechanism. At least in our set-up, even if financial variables do play a direct role in the way shocks are transmitted to the economy, they little modify the reaction of output and inflation to a monetary policy shock.

Below, we show that credit frictions are important in two additional respects. First, they modify the objective of monetary policy compared with the case of frictionless financial markets. Second, they affect the conduct of monetary policy when shocks which affect the macroeconomy originate in financial markets.

4. Second-order Welfare Approximation

Following Woodford (2003), we obtain a policy objective function by taking a second-order approximation of the utility function of the economy's representative agents. As our economy is populated by households and entrepreneurs, the policy objective function will be a weighted average of the (approximate) utility functions of these two agents. The approximation to the objective function takes a form which nests the one in the benchmark NK model (Woodford, 2003) as a special case.

Under the functional form for household's utility defined above, online Appendix F shows that the present discounted value of social welfare can be approximated by

$$W_{t_0} \simeq \varsigma c^{1-\sigma^{-1}} \left[\varkappa - \frac{1}{2} \mathrm{E}_{t_0} \sum_{t=t_0}^{\infty} \beta^{t-t_0} L_t \right] + t.i.p., \tag{17}$$

where ς is the weight assigned to households in the welfare function, *t.i.p.* denotes terms independent of policy and

$$L_t \equiv \kappa_\pi \pi_t^2 + \frac{1}{2}\frac{Y}{c}\varphi \tilde{Y}_t^2 + \frac{1}{2}\sigma^{-1}\tilde{c}_t^2 - \sigma^{-1}\frac{e}{c}\hat{c}_t^e \hat{e}_t + \frac{e}{c}\left(1 - \frac{1-\varsigma}{\varsigma}c^{\sigma^{-1}}\right)\left(\hat{e}_t + \frac{1}{2}\hat{e}_t^2\right). \tag{18}$$

$\hat{e}_t$ is log-entrepreneurial consumption (in deviation from the steady state) and $\varkappa$ and κ_π are parameters defined in online Appendix F.

The first three terms in (18) are common to the NK model. Social welfare decreases with variations of inflation around its target and of the output gap around its (non-zero) steady-state level. The first reason for disliking variations in the output gap is that households wish to smooth their labour supply. The second reason is that households also wish to smooth consumption over time. Unlike in the benchmark NK model, the consumption smoothing motive only applies to households' consumption, $\tilde{c}_t^2$, rather than to total output, because entrepreneurs are risk neutral and thus indifferent about the timing of their consumption.

The main difference relative to the benchmark NK model with frictionless financial markets is in the additional terms now appearing in the welfare approximation.

The term which is proportional to $\hat{c}_t^e \hat{e}_t$ contributes positively to welfare. The presence of this term is related to households' consumption smoothing motive (this term would disappear if households utility were linear, that is, when $\sigma^{-1} = 0$). Under an output expansion induced by a technology shock, an increase in entrepreneurial consumption absorbs aggregate resources and thus contributes to smooth the path of households' consumption over time.

The last term in (18), which is proportional to $\hat{e}_t + \hat{e}_t^2/2$, has an ambiguous impact on welfare, depending on whether the weight ς is larger or smaller than a threshold $\varsigma = (1 + c^{-\sigma^{-1}})^{-1}$. To understand this term, note first that $\hat{e}_t + \hat{e}_t^2/2$ is simply the second-order expansion of entrepreneurial consumption around its log level. If the weight of households in welfare were $\varsigma = 1$, then this term would have an unambiguously negative impact on welfare: entrepreneurial consumption represents a loss of resources from the perspective of households. In the opposite case in which households' weight in welfare were $\varsigma \to 0$, this term would contribute positively to welfare: more entrepreneurial consumption is unambiguously welfare improving from the perspective of entrepreneurs. For intermediate values of ς, the effect of entrepreneurial consumption on social welfare can be either positive or negative and it is negative when households' interests prevail – that is, when ς is larger than the aforementioned threshold, $\varsigma > (1 + c^{-\sigma^{-1}})^{-1}$. Note that the threshold would simply be $\varsigma = 1/2$ if households were risk neutral, that is, if $\sigma^{-1} = 0$.

The term proportional to $\hat{e}_t + \hat{e}_t^2/2$ in (18) highlights the potential redistributional effects of policy in our model. This term will also tend to be dominant from a quantitative perspective, because $\hat{e}_t$ is of first order. Depending on the exact weight of households in welfare, our model can therefore generate very different optimal policy responses to shocks. Given the arbitrariness of the choice of ς, in our numerical results we focus on the benchmark case in which the redistributional incentives of optimal policy are neutralised.[9]

The particular weight which achieves this objective is $\varsigma = (1 + c^{-\sigma^{-1}})^{-1}$.[10] As a result, first-order terms disappear entirely from social welfare. Under this special weight ς, the loss function simplifies to

$$L_t \equiv \kappa_\pi \pi_t^2 + \frac{1}{2}\frac{Y}{c}\varphi \tilde{Y}_t^2 + \frac{1}{2}\sigma^{-1}\left[\frac{Y}{c}\tilde{Y}_t - \frac{e}{c}\left(\hat{R}_t + \delta_3\hat{\Delta}_t + \hat{\tau}_t\right)\right]^2 - \sigma^{-1}\frac{e}{c}\frac{Y}{c}\hat{Y}_t^e\left(\hat{R}_t + \delta_3\hat{\Delta}_t\right), \quad (19)$$

where (9) and (10) were used to write entrepreneurial consumption in terms of the nominal interest rate $\hat{R}_t$, the credit spread $\hat{\Delta}_t$ and the exogenous shock $\hat{\tau}_t$. This expression allows us to perform a complete derivation of optimal policy using the linearised policy (11)–(13).

Compared with the case of the standard NK model, the novel terms in expression (19) are those with a coefficient proportional to e/c (note that these terms vanish when entrepreneurs disappear from the economy and $Y = c$) plus the last term.

These novel terms include first, within square brackets, elements proportional to the squared nominal interest rate and the squared loan-deposit rate spread. Hence, the presence of asymmetric information in the economy introduces directly both an interest rate smoothing and a 'spread smoothing' motive for optimal policy. At the same time, these terms are relatively small in our calibration, where households' consumption takes up the lion's share of output. Under normal circumstances, therefore, the interest rate smoothing concern is unlikely to be predominant compared to the objective of maintaining price stability.

The term within square brackets in (19) also includes a number of cross products between endogenous variables. More specifically, a planner would be averse to a

[9] We explore the robustness of our results to this assumption in the next Section.
[10] In our calibration, this yields a weight for households $\varsigma = 0.55$.

positive covariance between the nominal interest rate and the loan-deposit rates spread. A high covariance would increase the volatility of entrepreneurial consumption, with negative spillovers on households' consumption-smoothing motive. The planner would however not be averse to a positive covariance between, on the one side, the output gap, and, on the other side, either the interest rate or the loan-deposit rates spread. A negative output gap, for example, would be welfare improving if accompanied by a fall in the policy interest rate, such that only entrepreneurial consumption would suffer from the reduction in output, while households' consumption would remain unchanged.

The last term in (19) shows that increases in the nominal interest rate and in the credit spread have a positive effect on welfare, if they are accompanied by an increase in the efficient level of output – for example, an increase in productivity. The reason is that households are willing to reap the benefits of the higher productivity on real wages but wish to smooth their consumption pattern over time. Higher firms' profits and entrepreneurial consumption at a time of high productivity help to achieve the latter objective.[11]

Note that expression (19) is purely quadratic. Under this assumption, optimal policy can be studied using the linear system in (11)–(13).

5. Extensions

Below, we use our benchmark model under the central bank loss function in (19) to derive numerical results on the optimal monetary policy.

There are two key assumptions underlying this analysis. First, we have assigned an arbitrary weight to households in social welfare. Second, firms do not use profits to accumulate net worth, which is exogenously given. To evaluate the relevance of these assumptions for our results, in this Section we study two alternative versions of the model.

Concerning the specific weight on households' consumption in the social welfare function, ς, the value which eliminates linear terms from the second-order welfare approximation under our calibration is approximately one half. This does not appear to be an unreasonable benchmark but we have no hard evidence to validate it empirically. Our welfare results are obviously sensitive to changes in ς, because of the already emphasised quantitative relevance of linear terms in (18). As a robustness check, we therefore consider the case in which, as in the standard NK model, entrepreneurs become completely irrelevant from a welfare perspective. This allows us to study whether the different policy implications of our model, compared with the NK case, are mostly driven by the different economic structure *per se*, rather than by the combination of differences in both structure and welfare functions. To construct the

[11] In our derivations, gaps are defined in terms of deviations from the efficient equilibrium – an equilibrium where all nominal and real frictions are absent. An alternative possibility would be to write all variables in deviation from the 'natural' equilibrium – an equilibrium where nominal frictions (price rigidities and cost channel) are absent but real frictions (agency costs) are still present. It turns out, however, that in our economy, monetary policy can do better than implementing the natural equilibrium. Consistently with general second-best results, our numerical analysis below shows that optimal policy would choose not to maintain price stability at all times.

case in which entrepreneurs are irrelevant from a welfare perspective, we follow De Fiore *et al.* (2011) and consider a variant of our model (which we call 'DTnoE', fully described in online Appendix G) in which entrepreneurial consumption can be taxed at a constant rate ζ. We then focus on the limiting case in which ζ becomes arbitrarily large, so that entrepreneurial consumption approaches zero.[12] As $e \rightarrow 0$, (19) becomes

$$L_t \equiv \kappa_\pi \pi_t^2 + \frac{1}{2}\left(\varphi + \sigma^{-1}\right)\tilde{Y}_t^2. \tag{20}$$

The loss function collapses to the quadratic expression typically obtained in the optimal monetary policy literature in models with perfect financial markets. It corresponds to setting a unitary weight on households' utility in the welfare function of the social planner.

In the second extension of the model, we relax the assumption that firms' internal funds, τ_t, are exogenous. The exogeneity of internal funds is a convenient assumption because it enables us to obtain a reduced form of the system which we can easily relate to the benchmark NK model, and to derive a simple quadratic approximation to the welfare function. Nonetheless, it comes at the cost of leading us quite far away from the standard formulation of models in the 'financial accelerator' literature, where the accumulation of firms' internal funds is one important component of the transmission mechanism. We therefore study also an extension of our model (denoted 'DTacc', which is also fully described in online Appendix G) where firms can accumulate nominal internal funds. As we assume that entrepreneurial consumption is completely taxed away, these funds equate the profits (after repayment of the debt) realised by firms in the previous period.

In this case, the accumulation equation for aggregate internal funds is given by

$$\pi_t b_t = (1-\gamma)\left[1 + \frac{\mu\phi(\bar{\omega}_t)}{f_{\bar{\omega}}(\bar{\omega}_t)}\right]^{-1} R_t b_{t-1}\varepsilon_t^{\upsilon}, \tag{21}$$

where γ is the probability of death for entrepreneurs, $b_t = B_t/P_t$ is the real value of firms' internal funds, and ε_t^{υ} is a serially uncorrelated shock which alters the value of accumulated internal funds at the time when they are used for production.[13]

6. Optimal Monetary Policy: A Numerical Analysis

We characterise optimal monetary policy under commitment numerically. All our results are based on the timeless perspective, as in Woodford (2003).[14]

[12] Note that the terms involving entrepreneurial consumption in the welfare approximation – (18) – are all proportional to the steady-state level of entrepreneurial consumption. They therefore tend to zero at the same speed as entrepreneurial consumption. In other words, they become negligible once entrepreneurial consumption becomes negligible as a share of output. Hence, there are no discontinuities in the equilibrium as $\zeta \rightarrow \infty$. We can verify this result numerically. The impulse responses in Figure 4 are equivalent, if we assume a high tax rate ($\zeta = 150$) such that entrepreneurial consumption is 0.1% of output, or in the limit where $\zeta \rightarrow \infty$.

[13] This is akin to the capital quality shock used by Gertler and Kiyotaki (2010) and others.

[14] Our numerical results are based on a full, second-order approximation of the policy equations, which we perform using Dynare. We compute the first-order conditions of the welfare maximisation problem of the policy maker using Giovanni Lombardo's lq_solution routine available at http://home.arcor.de/calomba/symbsolve4_lnx.zip.

Figure 3 displays impulse responses to a negative technology shock when monetary policy is set optimally. Once again, we compare the responses in the credit channel model – that is, our benchmark model under the central bank loss function in (19) – to those in the model of the cost-channel, and in the standard NK model. Here, we also display results based on the extension of our model with endogenous accumulation of net worth.[15]

As is well known (Woodford, 2003), optimal policy would ensure complete price stability and full stabilisation of the output gap in the NK model. To achieve this outcome, the policy interest rate would rise on impact and then return slowly to the baseline.

This response of the policy interest rate is similar across models but inflation and the output gap move in different directions. In the model of the cost channel, inflation rises reflecting the increase in marginal costs due to the higher interest rate. Higher marginal costs depress the output gap, which remains negative throughout the adjustment period.

In our benchmark model, credit, leverage and bankruptcies fall. The resulting reduction in the spread more than compensates the increase in the policy rate, leading

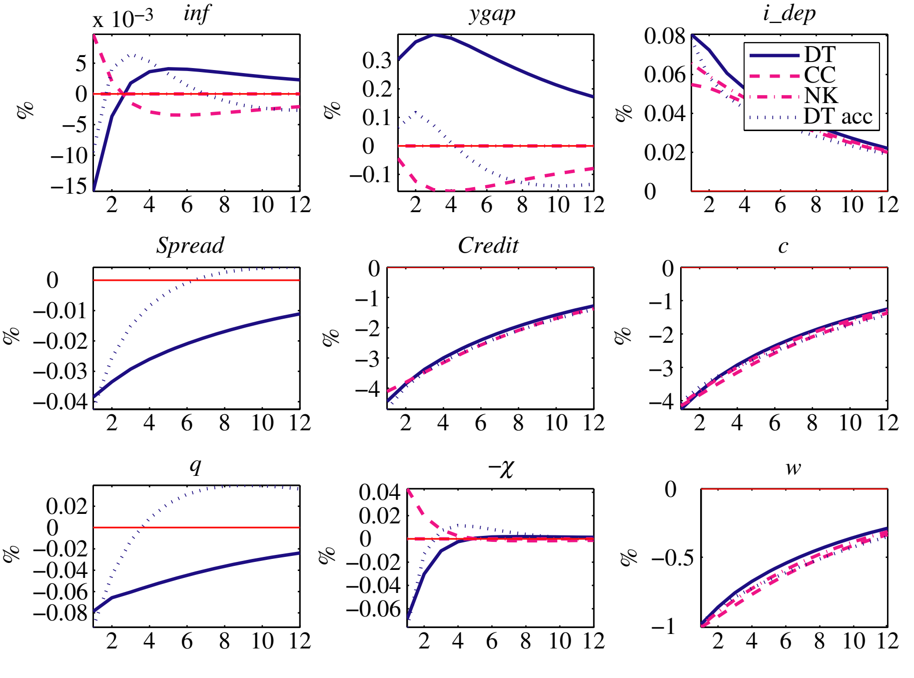

Fig. 3. *Impulse Responses to a Technology Shock under Optimal Policy* (see Notes to Figure 1)

[15] In all cases, we assume the existence of a steady-state subsidy, which eliminates first-order terms in output from the second-order expansion of individuals' utility. The subsidy is slightly different in the three cases: it is equal to $\chi/(\chi - 1)$ in the new-Keynesian model, $R\chi/(\chi - 1)$ in the cost-channel model and $q\chi/(\chi - 1)$ in our model and in the model with endogenous internal funds. For each of the models we analyse, 'optimal policy' refers to the solution of the Ramsey problem (under the timeless perspective) in the same model. In the model with endogenous accumulation of internal funds, we keep σ_ω fixed and adjust the probability of death of entrepreneurs γ so as to ensure that the steady-state values of credit spreads and bankruptcy rates are unchanged from the benchmark model.

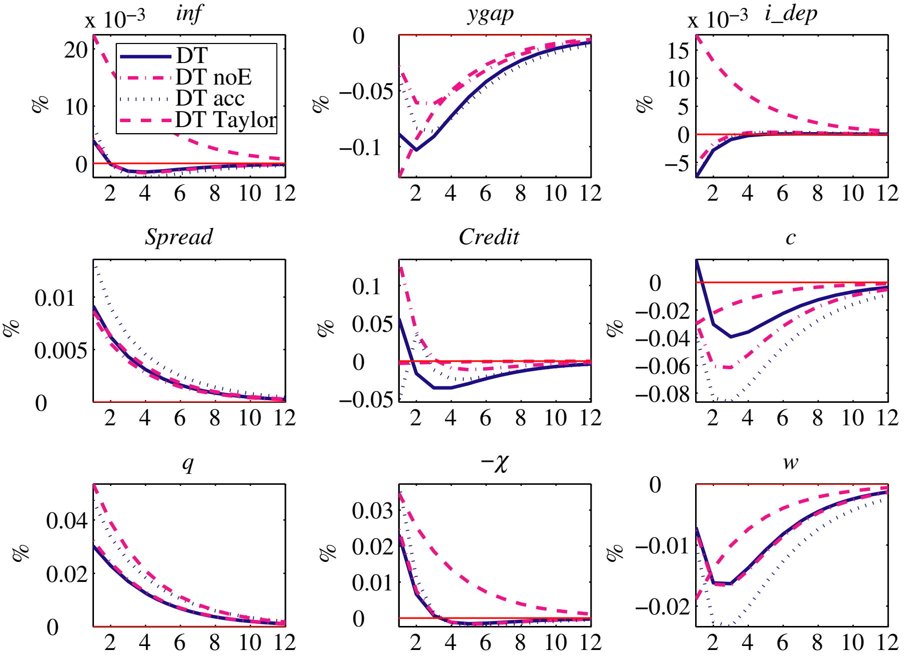

Fig. 4. *Impulse Responses to a Net Worth Shock: Taylor Rule Versus Optimal Policy* (see Notes to Figure 1)

both to a lower inflation and a slightly lower financial markup $\hat{q}_t$. The reduction in this markup implies that real wages fall less than one-to-one with the shock.

On impact, the fall in credit, leverage and bankruptcies is very close to the benchmark case in the model with accumulation of internal funds. Over time, however, firms accumulate net worth, so the adjustment path for these variables is faster. The output gap raises less on impact and then becomes negative before returning to the steady state. Nevertheless, the responses of consumption, credit, real wages and the policy rate are quantitatively very similar to the benchmark case.

Figure 4 shows the impulse responses to a shock that reduces the value of firms' internal funds under optimal policy. This shock is representative of a broader array of 'financial shocks' which could be defined in our model, notably shocks to the monitoring costs μ, or to the standard deviation of idiosyncratic shocks, σ_ω. The impulse responses in our benchmark model are compared with the responses under the Taylor rule and to the optimal responses which would be obtained in the two extensions described in Section 5.[16] Note that in the case with endogenous accumulation of net worth the shock is to ε_t^{v} in (21) and it is calibrated to generate a similar impact fall in net worth as the fall in $\hat{\tau}_t$ in the benchmark model.

In our benchmark model, the reduction in the value of internal funds, which is assumed to be persistent, acts like a classical cost-push shock: it tends to generate

[16] The model without entrepreneurial consumption requires a small adjustment in the calibration of τ and σ_ω to ensure that the steady-state values of credit spreads and bankruptcy rates are unchanged from the benchmark model.

inflationary pressures and a negative output gap. Inflationary pressures follow from the increase in leverage after the shock, which drives up the loan-deposit rate spread and firms' marginal costs. The negative output gap is due to the increase in the markup q_t, which also follows from the increase in marginal costs and exerts downward pressure on wages. Households react to the lower wage rate with a reduction in both their labour supply and in their demand for consumption goods, inducing a persistent reduction in the output gap.

The policy response under a Taylor rule is to increase interest rates in the face of the rising inflationary pressure. In spite of this policy response, inflation rises by 1% point, partly due to the cost-channel effect of the nominal interest rate. The reduction in $\hat{\tau}_t$ also leads to an increase in bankruptcy rates and a small fall in the amount of credit.

Hence, contrary to the case of the technology shock, bankruptcy rates and the credit spread move anti-cyclically in response to a financial shock.

Compared with the responses under the Taylor rule, those obtained under optimal policy are striking because the policy interest rate moves in the opposite direction. In spite of the inflationary pressure, interest rates are immediately cut and stay low for approximately one year. The main reason for this policy response is that the financial shock is inefficient; hence, the fall in the output gap is entirely undesirable. The marked expansion in monetary policy is aimed at smoothing households' consumption path after the shock. Consumption of entrepreneurs falls so sharply due to the fall in net worth that households' consumption can increase slightly on impact. All in all, inflation is stabilised better than under the Taylor rule, also thanks to a smaller upward pressure through the cost channel.

Impulse responses are very similar in the model where entrepreneurial consumption is taxed away. The main difference is that the sharp fall in entrepreneurial profits after the shock does not affect aggregate variables (because it does not lead to a fall in entrepreneurial consumption). Output falls less than in the benchmark case and households' consumption does not increase on impact.

Unsurprisingly, the case with endogenous accumulation of net worth is closest to the version of our model in which entrepreneurial consumption is fully taxed. The impact fall in consumption and output is similar across these two cases. However, the financial markup increases more to finance the accumulation of net worth, when the latter is endogenous. As a result, wages fall more and so do consumption and output along the adjustment path. Once again, the impulse responses of the nominal interest rate and inflation are virtually identical to the benchmark case.

7. Conclusions

Using a small, microfounded model with nominal rigidities and credit frictions, we analyse the implications of financial market conditions on macroeconomic dynamics and on optimal monetary policy.

In our simple set-up, it is possible to characterise analytically the linearised aggregate relations of the model and to obtain an approximate welfare criterion consistent with the microfoundations of the model. Our results show that, in general, monetary policy ought to pay attention to the evolution of financial market conditions, as captured for example by changes in credit spreads. On the one hand, these changes matter because they affect firms' marginal costs and therefore have an impact on output and inflation. On the other hand, they may matter because of their impact on welfare.

In our numerical analysis, we find that despite the presence of cost-push factors introduced by financial market frictions, optimal monetary policy does not produce substantial deviations from price stability. Nevertheless, our results suggest that there might be good reasons for a central bank to react with an aggressive easing to an adverse financial shock. Those types of shocks create an inefficient recession, whose negative consequences on consumption can be reduced.

Our results should help to improve our understanding of the determinants of optimal policy decisions in models with credit frictions. We have highlighted the key trade-offs faced by monetary policy when setting interest rates in response to financial shocks. Another dimension of optimal policy which we have not explored in this article is the potential role for 'non-standard' measures, which have been used extensively by central banks in reaction to the financial crisis of 2007–8. In our model, policies that help firms to deleverage more quickly after a fall in their net worth are potentially beneficial. It remains to be seen whether these benefits are sufficiently large to justify sizeable public interventions.

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Additional Supporting information may be found in the online version of this article:

Appendix A. The Financial Contract.
Appendix B. Price Setting.
Appendix C. The System in Reduced Form.
Appendix D. Gaps Relative to the Efficient Equilibrium.
Appendix E. Case with Zero Monitoring Costs.
Appendix F. Welfare Approximation.
Appendix G. Robustness.

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